

Joshua Kim

Curriculum Vitae

Department of Computer Science
University of Texas at Austin

Research Interests

- Computer architecture and hardware-software co-design for datacenter efficiency.
- Acceleration of complex problems using heterogeneous architectures.
- Compiler tools that offload resource management and parallel code from programmers.

Education

- 2025–Present **Ph.D Computer Science**, *University of Texas at Austin*
Advisors: Prof. Jovan Stojkovic, Prof. Christopher J. Rossbach
Topic: Designing Processor Architectures for Datacenter Servers
- 2021–2025 **Bachelor of Science, Computer Science & Mathematical Sciences**, *Carnegie Mellon University, (Dual Degree)*
Thesis: Towards a Scalable Multi-GPU System for Encrypted AI **GPA:** 3.73/4.0

Conference Publications

- C1. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic.
“Phaseweave: Phase-aware Execution on Heterogeneous Chiplet Architectures for Datacenters”, *To Appear In Proceedings of the 53rd International Symposium on Computer Architecture (ISCA)*, June, 2026.

Workshop and Other Publications

- W2. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic.
“Hunting for Offload: Automated Discovery of Acceleratable Code in Datacenters”. *In Architecture 2.0: Workshop on AI for Computing Systems Design with ASPLOS*, March 2026
- W1. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic.
“Lightweight AI for Efficient Resource Management in Heterogeneous-Core Architectures”. *In Architecture 2.0: Workshop on AI for Computing Systems Design with ASPLOS*, March 2026

Scholarships & Awards

- 2026 **Student Travel Grants**, ASPLOS’26
- 2025 **Allen Newell Award for Excellence in Undergraduate Research**, Carnegie Mellon University
- 2024-2025 **CRA Outstanding Undergraduate Research Awards - Honorable Mention**, Computing Research Association

Research Experience

Aug 2025 - **Research Assistant, UT Austin**, *Advisors: Prof. Jovan Stojkovic, Prof. Christopher J. Rossbach*, Designing Processor Architectures for Datacenter Servers
Present
May - Aug 2026 **Intern, Azure System Research, Microsoft Research**, *Advisors: Dr. Chaojie Zhang, Dr. Íñigo Goiri*, Fungibility of Training and Inference, Redmond, WA.
Aug 2024 - **Undergraduate RA, Carnegie Mellon University**, *Advisor: Prof. Dimitrios Skarlatos*, Towards a Scalable Multi-GPU System for Encrypted AI
Aug 2025
Nov 2023 - **CS Academy, Carnegie Mellon University**, *Advisor: Prof. David Kosbie*
May 2024 Clustering in Plagiarism Detection

Technologies

Languages: C, C++, CUDA, Python, Java, OCaml, SML, x86/64 Assembly, SQL

Hardware: CPU, GPU, AWS Trainium

Architecture simulators: SST, QEMU, Pin

Benchmarking Technologies: DCPperf, Intel Perfspect, VTune, CMT-CAT, Nvidia-SMI, Nsight Systems, McPAT

Selected Projects

1. Designing Processor Architectures for Datacenter Servers at Hyperscale

Modern datacenters run workloads with varying characteristics, and with the rise of the 'microservice' paradigm, we see differences in the components of a single application as well - resulting in sub-optimal performance and resource management. We propose a heterogeneous chiplet-based manycore architecture for datacenter servers, where each chiplet has different hardware configurations (e.g. cache sizes). When an application runs on the server, our system detects phase changes within an application and migrates the execution of the application to a chiplet that has the optimal hardware configuration for the current phase.

2. Scalable Multi-GPU System for Encrypted AI

Fully homomorphic encryption (FHE) is a cryptographic scheme that offers user data privacy by supporting computations directly on encrypted data - however, it suffers a severe performance overhead of four orders of magnitude compared to plaintext computations. To make FHE feasible in modern datacenters, we employ a scale-out approach on multi-GPU systems. We provide a GPU-targeting acceleration backend, along with a set of kernel-fusion, cross-device communication, and memory management techniques to address overheads unique to GPU implementations of FHE.

Teaching Experience

Undergraduate Teaching Assistant, Carnegie Mellon University

Fall 2024 – 15-213/513: Introduction to Computer Systems (*Head TA*)
Summer 2025
Spring 2025 15-418/618: Parallel Computer Architecture
Fall 2023 – 21-259: Calculus in Three Dimensions
Spring 2024